

TOROW Function in Excel

1. What is TOROW?

The TOROW function takes a range with many rows and columns, or a single column, and lays all its values out in a single row. It solves the problem of data that is arranged in a grid or a list when you need it side by side. A Data Analyst uses it to flatten tables and to turn column lists into rows, for example to create column headings for a summary table.

2. Syntax

`=TOROW(array, [ignore], [scan_by_column])`

Argument	Required?	What it accepts
array	Required	The range or array to convert into one row (for example A1:C3)
ignore	Optional	What to skip: 0 keep all (default), 1 ignore blanks, 2 ignore errors, 3 ignore blanks and errors
scan_by_column	Optional	FALSE (default) reads row by row, left to right. TRUE reads column by column, top to bottom.

For a grid with 1, 2, 3 in the first row and 4, 5, 6 in the second: the default gives 1, 2, 3, 4, 5, 6, and TRUE gives 1, 4, 2, 5, 3, 6. To use scan_by_column and keep the default for ignore, leave the middle argument empty: `=TOROW(A1:C3, , TRUE)`.

3. Simple Example

Turn a 3 × 3 grid into one row, ignoring blank cells:

	A	B	C	D	E	F	G
1	Apple	Fig	Lime				
2	Pear		Kiwi				
3	Plum	Date					
4	Apple	Fig	Lime	Pear	Kiwi	Plum	Date

Formula in A4 only: `=TOROW(A1:C3, 1)`
Excel reads row 1 (Apple, Fig, Lime), then row 2, then row 3, skipping the two blank cells (B2 and C3). The 7 values spill across A4 to G4. **Result: 7 values in one row**

4. Another Example

Collect the numbers from a grid and remove the errors:

	A	B	C	D
1	120	#N/A	80	
2	95	60	#N/A	
3	120	80	95	60

Formula in A3 only: `=TOROW(A1:C2, 2)`
The ignore value 2 tells Excel to skip errors. It reads row 1 (120, 80), then row 2 (95, 60), leaving out both #N/A cells. **Result: 120, 80, 95, 60.** Functions like SUM and AVERAGE can now work on this clean row.

5. When to Use It

- Flatten a multi-column table into one row
- Turn a vertical list into a horizontal row, such as column headings for a summary table
- Combine several columns of values into one row for other formulas
- Remove blanks and errors while reshaping data
- Prepare a row of items for functions such as UNIQUE or SORT

6. Important Things to Remember

- TOROW is a dynamic array function. You type one formula and the results spill to the right.
- By default, TOROW reads row by row. The third argument is scan_by_column: FALSE (default) reads by row, TRUE reads by column.
- By default, nothing is ignored, so blank cells appear in the result as 0 and errors are kept.
- Use 1 to skip blanks, 2 to skip errors, and 3 to skip both.
- The cells to the right of the formula must be empty, or Excel shows #SPILL!.
- TOROW is available in Microsoft 365 and Excel 2024, but not in Excel 2021 or earlier.

7. Common Mistakes

Mistake: Expecting blank cells to be skipped automatically, then finding 0 values in the result. **Fix:** Add the ignore argument: `=TOROW(A1:C3, 1)`.

Mistake: Using 1 to remove blanks, but still seeing #N/A in the result. **Fix:** Use 3 to ignore both blanks and errors, or 2 for errors only.

Mistake: Getting the wrong order because of the reading direction. **Fix:** Row by row is the default. Add TRUE as the third argument for column by column.

Mistake: Getting #SPILL! because cells to the right of the formula already contain data. **Fix:** Clear those cells or move the formula to an empty area.

8. Data Analyst Tip

Use TOROW to build headings that update by themselves. For example, `=TOROW(UNIQUE(A2:A20))` lists each unique item from a column across a row. You can use this as the column headings of a summary table, and the headings will grow or shrink with your data.